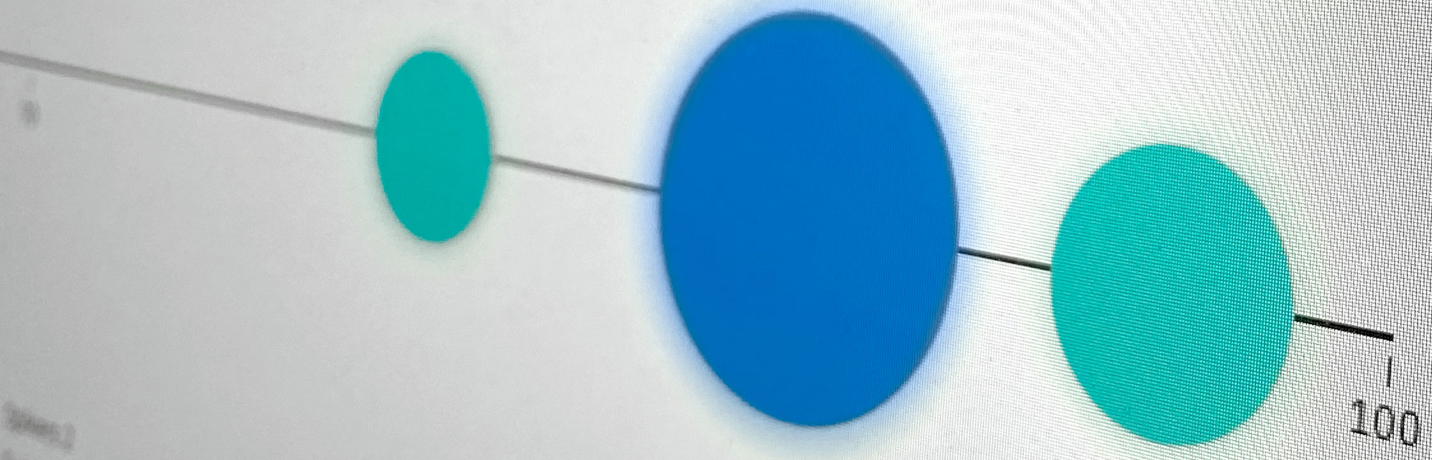
IBM Predictive Customer Analytics

Explore all customers

High attrition audiences

attrition Score 70 by amount of customers



Recent lists

Hipster Outreach Q2

323,442 Customers
$104 CLTV
48 Attrition score

Attrition, Hipsters, Cowbo
and 2 more rules.

Predictive Customer Analytics (PCA)

A cloud-based product for marketing analysts at a time when campaigns need to be personal and timely.

GOAL

The product needed to simplify and integrate predictive analytics into the core workflows of marketing analysts.

CLIENT AND SCOPE

The internal client was the IBM Analytics product division and I spent one year working on the product between 2015 and 2016.

ROLE

I was the primary product designer. My role was diverse and my most lasting contribution was defining the product's core user flow.

OUTCOMES

I brought the product from its initial user and market need framing to its limited released to clients.

How might we design a better way for marketing analysts to identify customers for a sales campaign?

IBM Analytics sought to contextualize their portfolio to meet niche market needs

IBM Analytics, a division of IBM's Software Group, relied on two products. SPSS was a statistic software suite focused on advanced analytics and Cognos was a business intelligence product focused on reporting. By 2015, IBM Analytics was ready to bring together the capabilities of both products to address a variety of market needs.

IBM Analytics recognizes design as a key ingredient to help marketing organizations

One need they identified was within marketing organizations. Customer expectations were driving companies to do more to satisfy customers. By making their marketing efforts personalized and time-sensitive, they could gain a competitive advantage. But problematically their organizational capabilities hindered customer centricity.

IBM Analytics believed they could address this gap, but understood colliding their products together would not be enough. They needed to leverage their capabilities in an experience that met the needs of actual people performing the work in a marketing organization. That is when the Predictive Customer Analytics (PCA) team called upon my design team for help.

PCA targets the frustrations of marketing analysts identifying people for campaigns

PCA's users, marketing analysts, work with campaign managers to produce lists of customers for marketing campaigns that's shaped both by data analysis and business requirements.

Frequently marketing analysts use spreadsheet-based products to target customers. These products tend to perform poorly with large sets of data, lack integration with marketing engagement systems, and cannot leverage techniques like predictive analytics. Further, marketing analysts are beholden to data scientists to create predictive models. Overall, today's organizations are limited in their ability to use predictive intelligence to create more targeted campaigns.

PCA seeks to democratize predictive analytics and integration

To improve this, marketing analysts need an experience that will allow them to have a robust and dynamic conversation with customer data to build lists that seamlessly integrate into the campaign workflow.

PCA's vision to think big and start small first focused on retaining customers

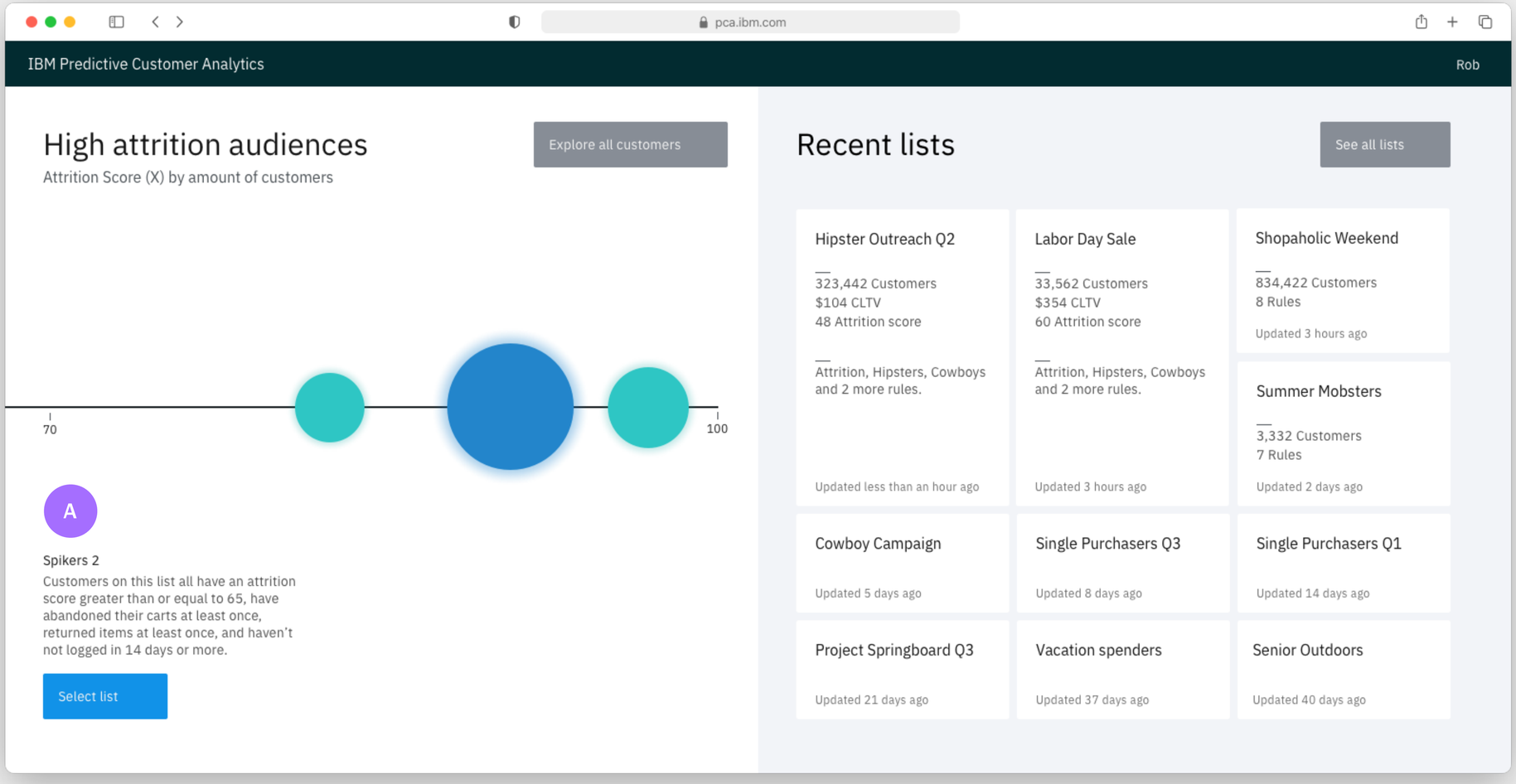
The first release of PCA would enable marketing analysts to:

- Identify audiences based on retaining customers with 2–3 times more accuracy than using their own intuition and without the help of a data scientist.
- See a real-time understanding customers with 5–7 descriptors in one view.
- Stream customer lists to marketing engagement systems without manual configuration.

Future releases of PCA would enable marketing analysts to:

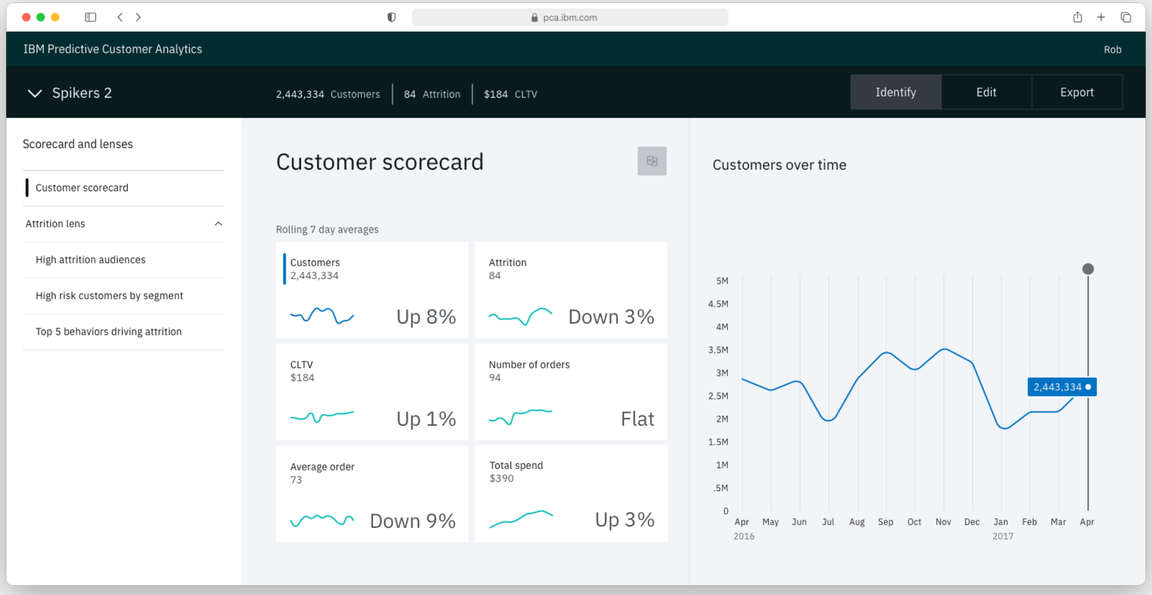
- Identify audiences based on engaging, growing, and attracting customers with 2–3 times more accuracy than using their own intuition and without the help of a data scientist.
- See a real-time understanding od customers with 8–12+ descriptors in one view.
- Track how lists change over time without having to remember important thresholds or inspect lists manually.

Begin analyses with starting points
Marketing analysts can jump start their analysis with lists generated by advanced analytic models (A).

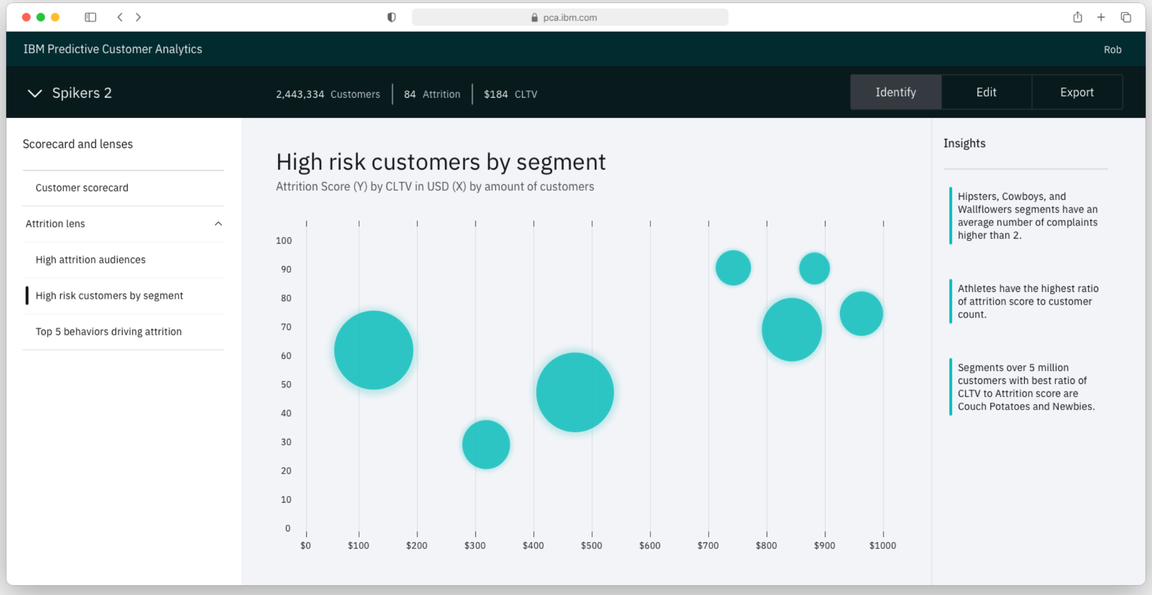


Monitor and identify customers

Using a scorecard and lenses, marketing analysts can monitor customer lists and identify customers with insights aided with visualizations.



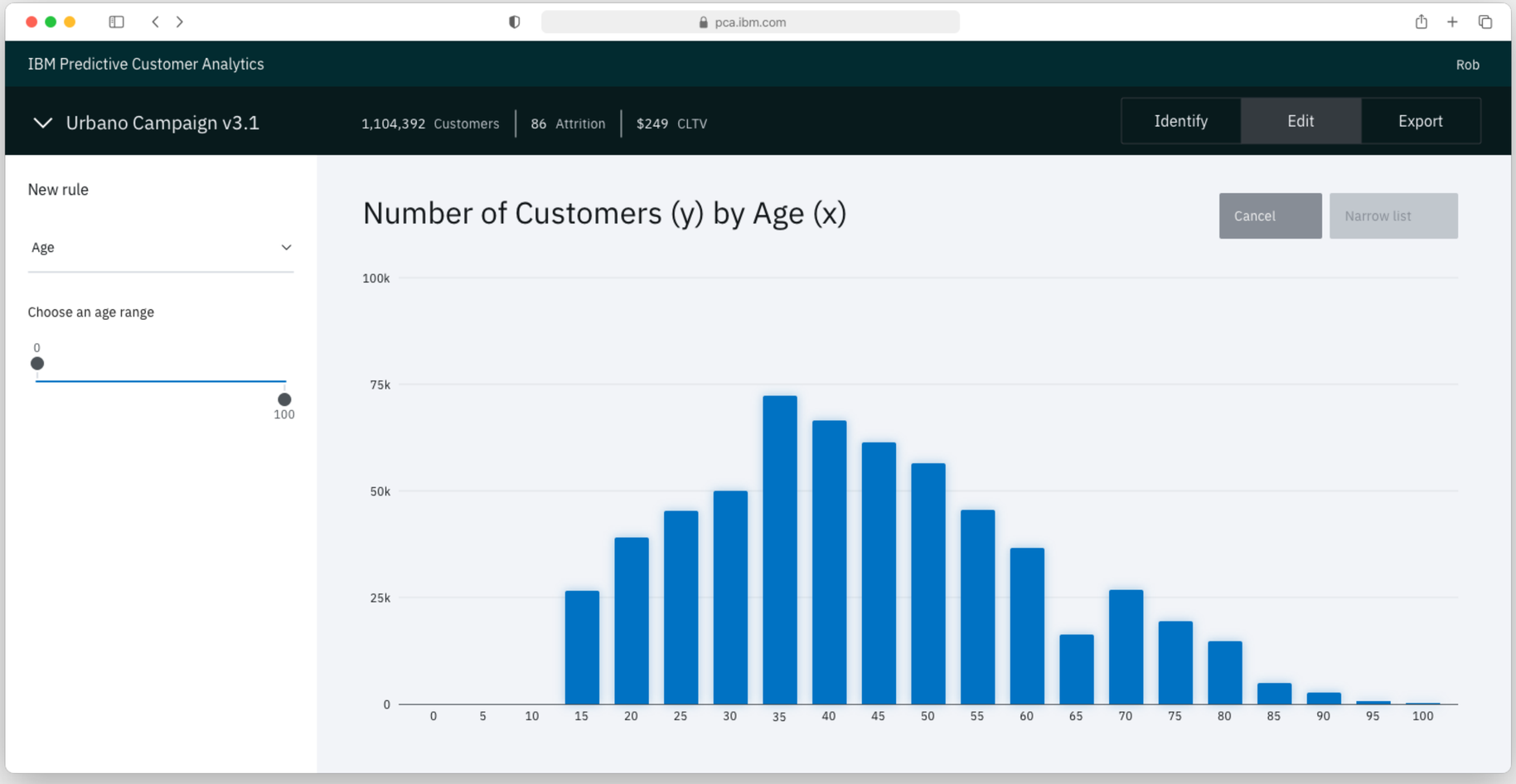
Customer scorecard



Section of the Attrition lens

Manually edit customer lists with rules

Marketing analysts can take full control over a list by manually shaping it to specific requirements of a campaign.



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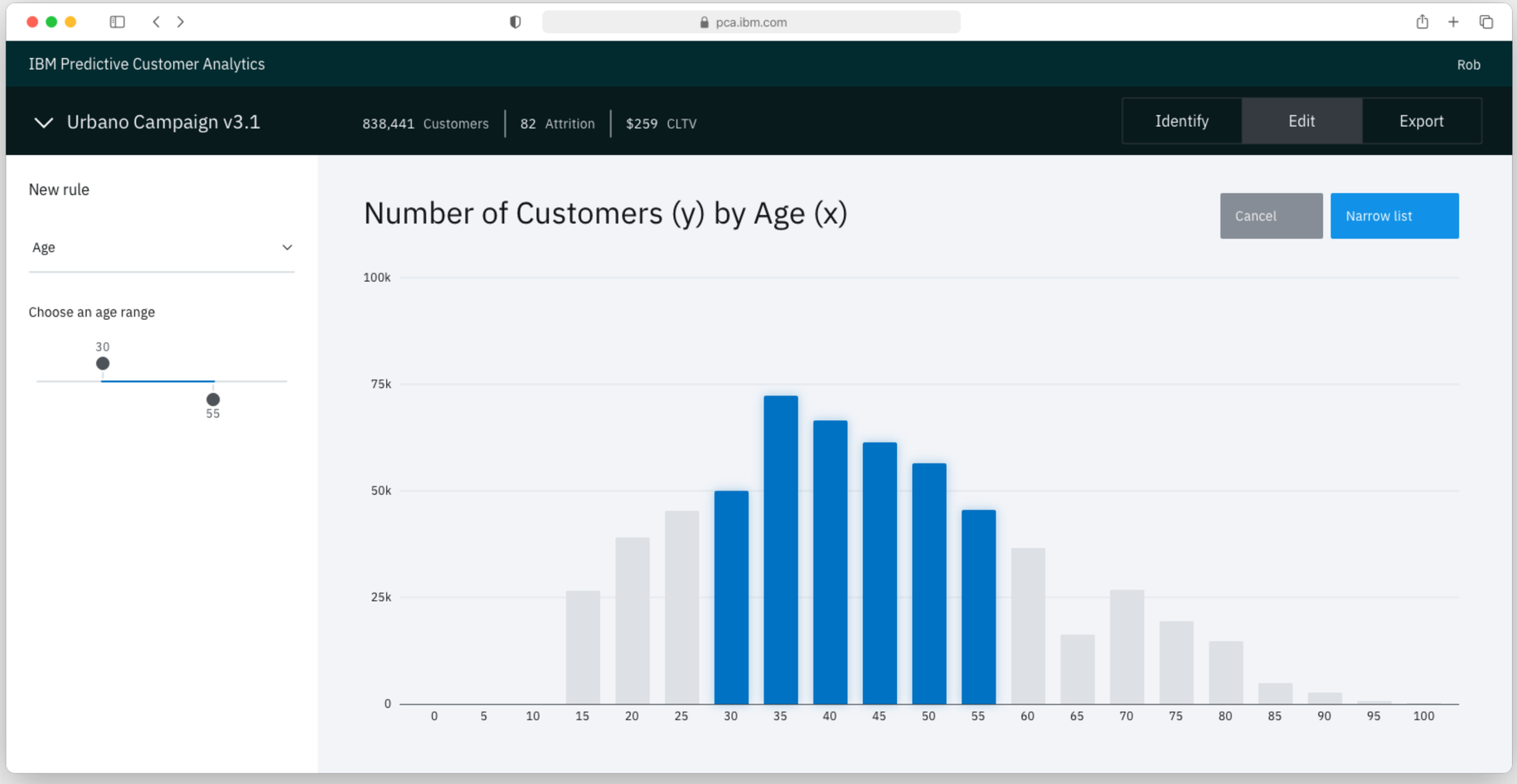

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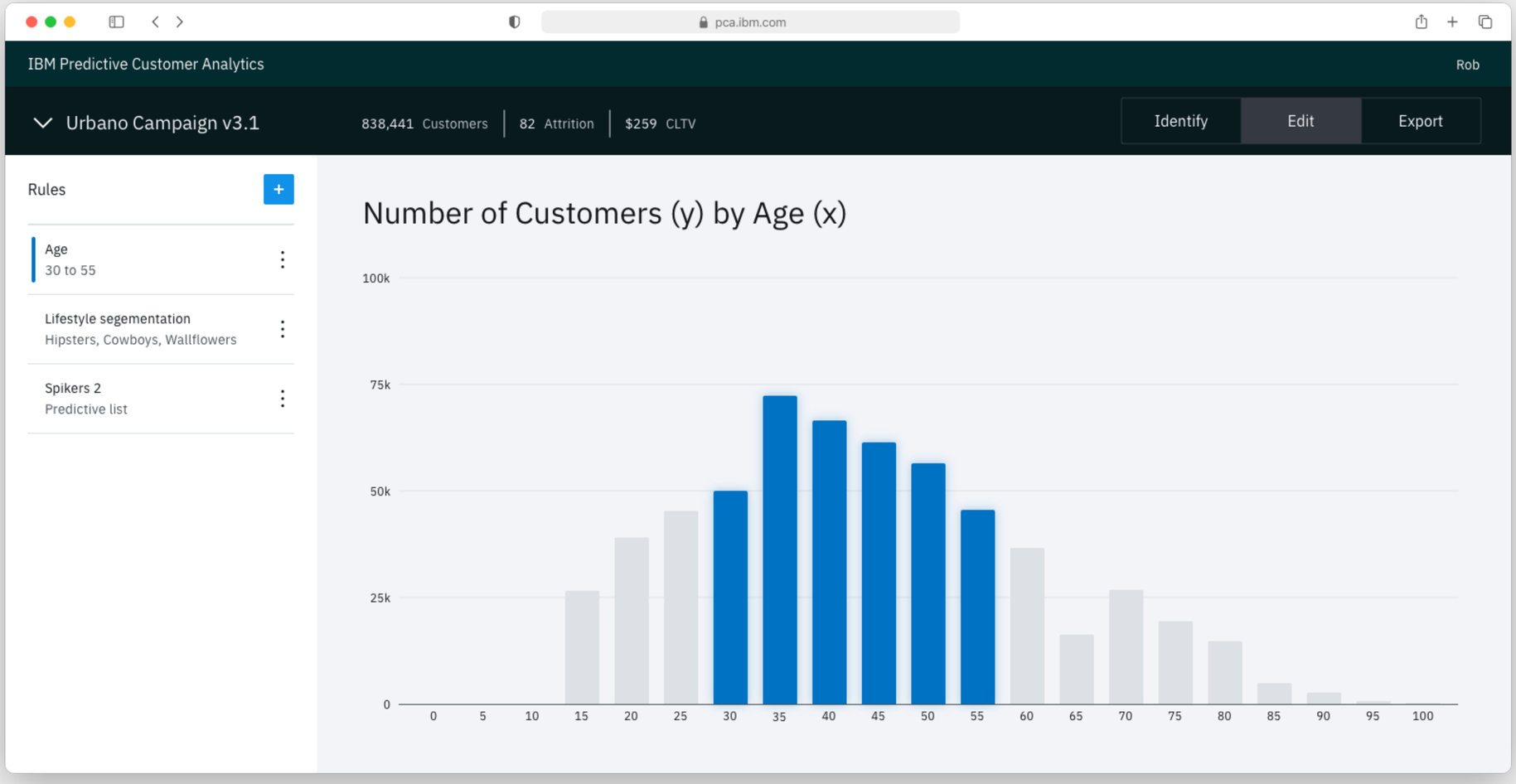
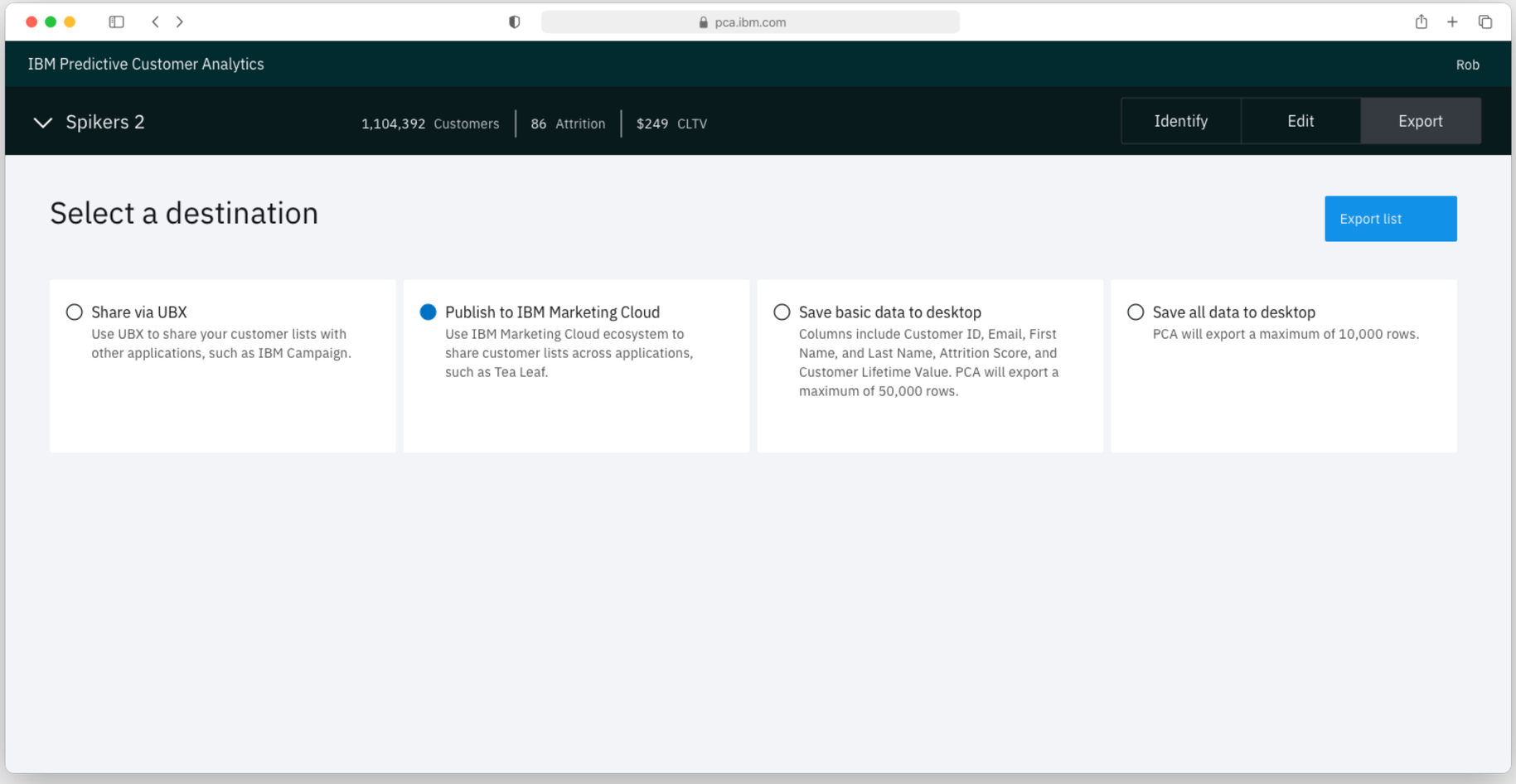
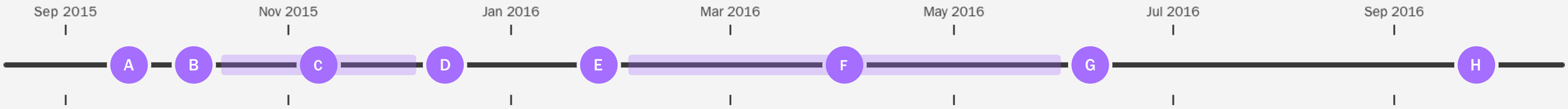


Export to many platforms

Marketing analysts can export or link customer lists to a number of formats and marketing systems.



PCA timeline
12 months, 2015–2016



(A) Framing workshop
The product team formed and design facilitated a workshop to align on the user, their problem, and PCA's goals.

(B) I join the team
The team provided me hills and I acclimated myself to the marketing analyst's knowledge domain.

(C) Approach playbacks
I presented concepts and to-be experiences that narrowed in on the core user experience for the application.

(D) Final approach playback
I designed and presented a user flow that aligned the team and moved the product into development.

(E) Application map breakthrough
During a workshop, I collaborated with product management and development to define the product's application map.

(F) Alpha and beta releases
I collaborated with development using an agile methodology to release the product's alpha and beta releases.

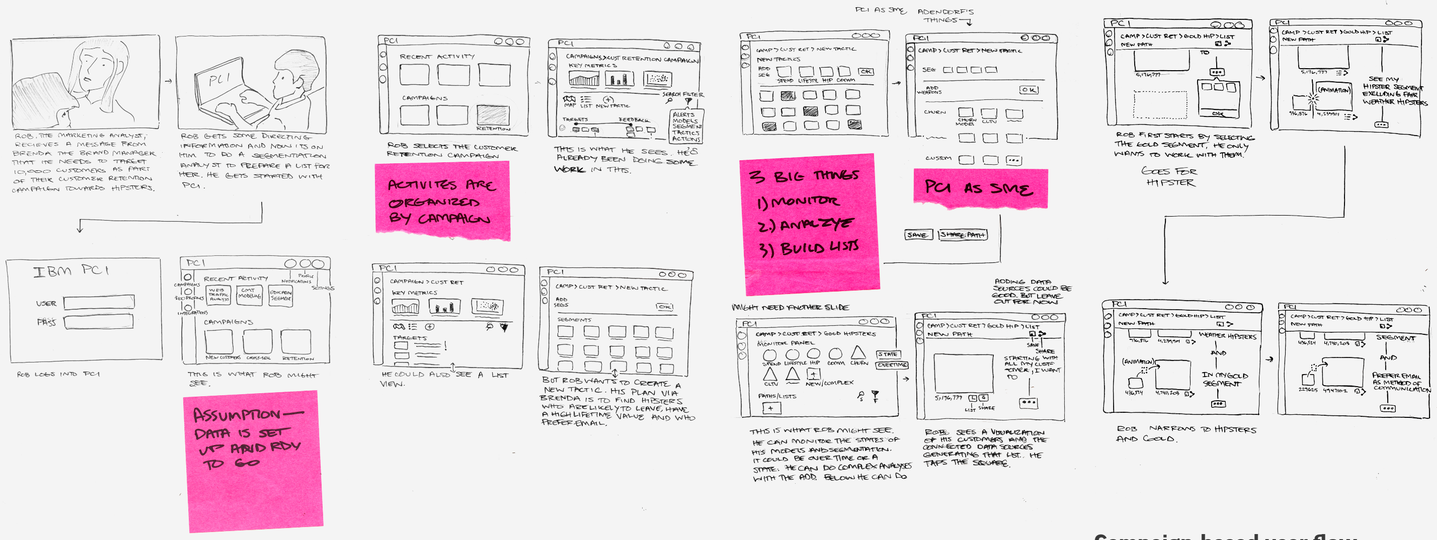
(G) I introduce the workspace concept
Presented with constraints and emerging ideas, I designed critical adjustments to the application's flow and interface.

(H) PCA moves to Commerice portfolio
IBM Analytics rationalizes PCA into the Watson Commerce portfolio and I use the opportunity to exit the team.

Defining PCA's core workflow involved exploring and presenting many concepts

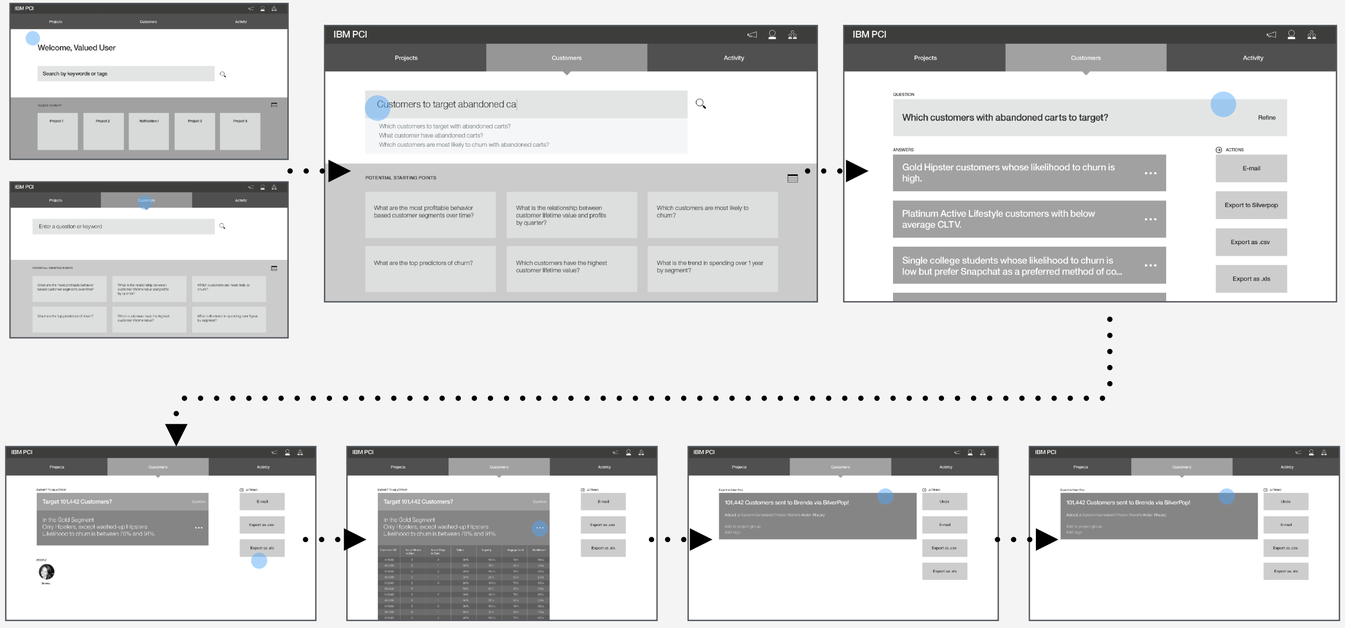
Arriving at initial end-to-end experience took several cycles of generating concepts and presenting them back to product management and engineering.

While none of these explorations shown here were successful, they helped identify and work through gaps in my team's understanding of marketing analysts and our vision for them.



Campaign-based user flow

Early designs referenced the campaign workflow, something we believed the user would be familiar with.



Question-based user flow

This mid-fidelity prototype shared at a playback demonstrates how an analyst primarily asks questions during their analysis.

I convinced stakeholders to evolve PCA's core workflow to match user needs

Stakeholders initially gravitated toward an application map that reflected a conventional file management system **(A)**. Concurrently, user research illuminated how analysts approached analysis.

My role was key in straddling the tension between championing user needs and managing the expectations of our business. Successfully, I challenged them to evolve the map so it better followed the needs of users.

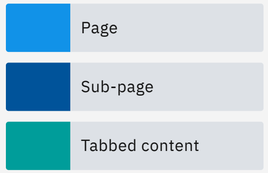
We landed on an application map with two connected content structures called a workspace **(B)**. One part allowed marketing analyst to access customer lists while the other allowed them to move bi-directionally through a data analysis workflow.

This structure facilitated the marketing analyst's cyclical workflow between understanding and shaping the list uncovered in our research.

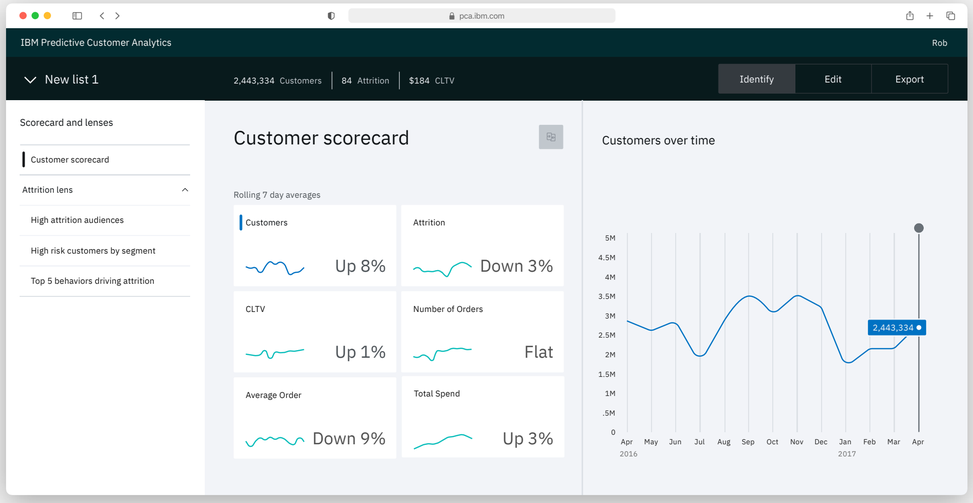
A Stakeholder's standard

B Design's vision

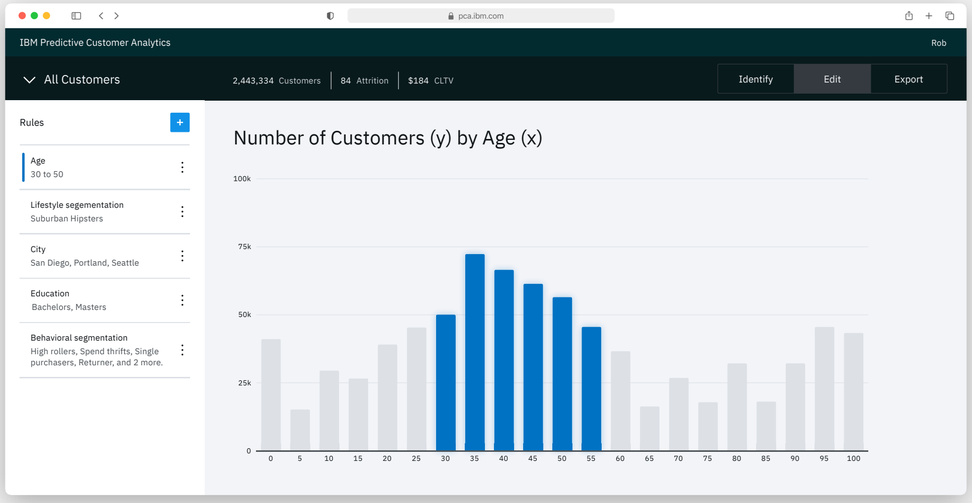
Key



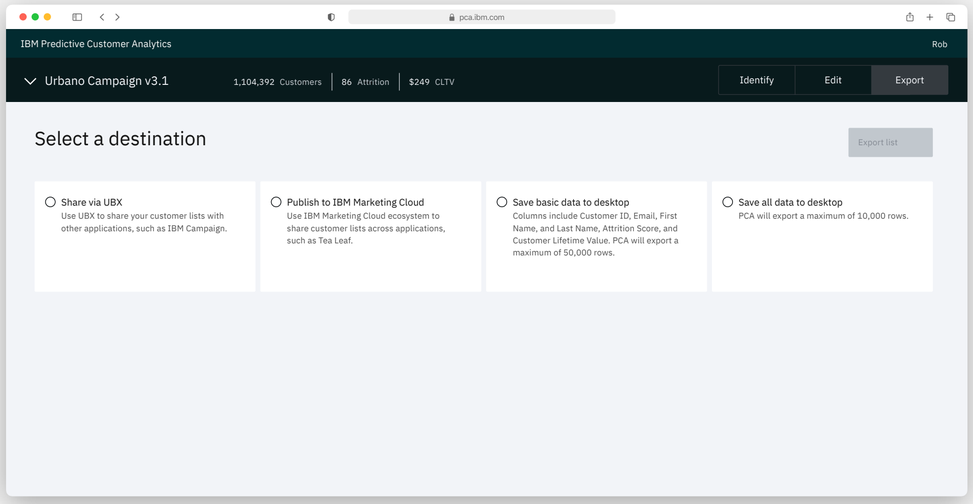
Workspace components



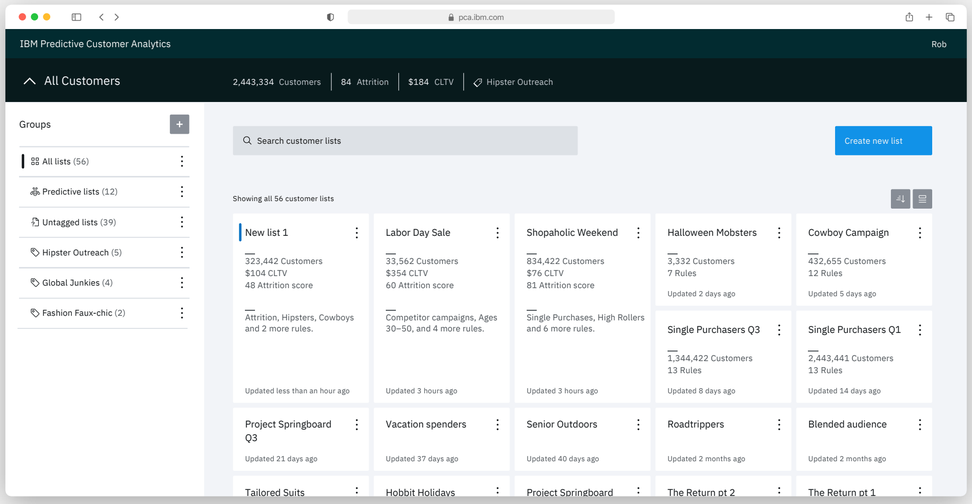
Identify in workspace



Edit in workspace



Export in workspace

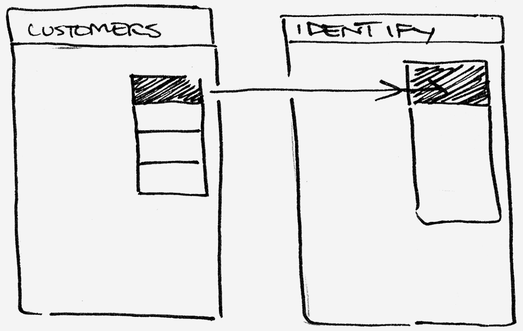


Lists in workspace

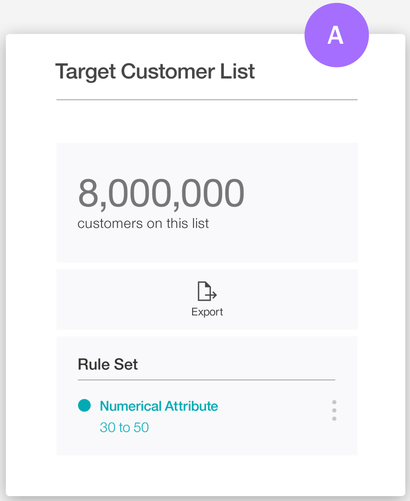
I helped users navigate a cyclical workflow by developing a component

In early designs it was unclear how marketing analysts could alternate between understanding customers and shaping a list. By the beta release, I designed a common user interface component, the Target Customer List (A), that displayed between these two tasks.

In later iterations, I nested the Target Customer List component within the global navigation (B). I also included key metrics about the list to provide a persistent reference point.

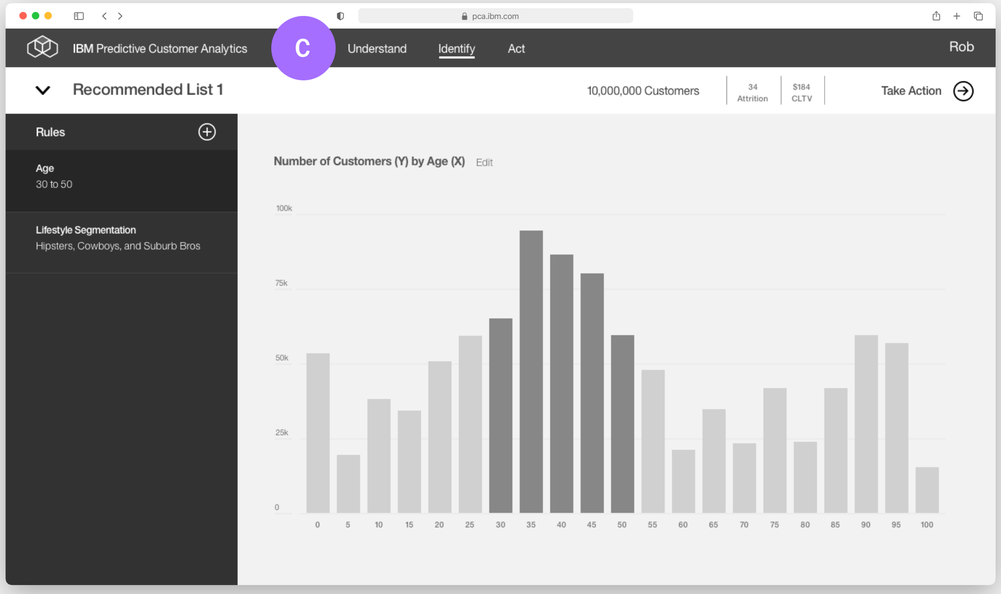


A



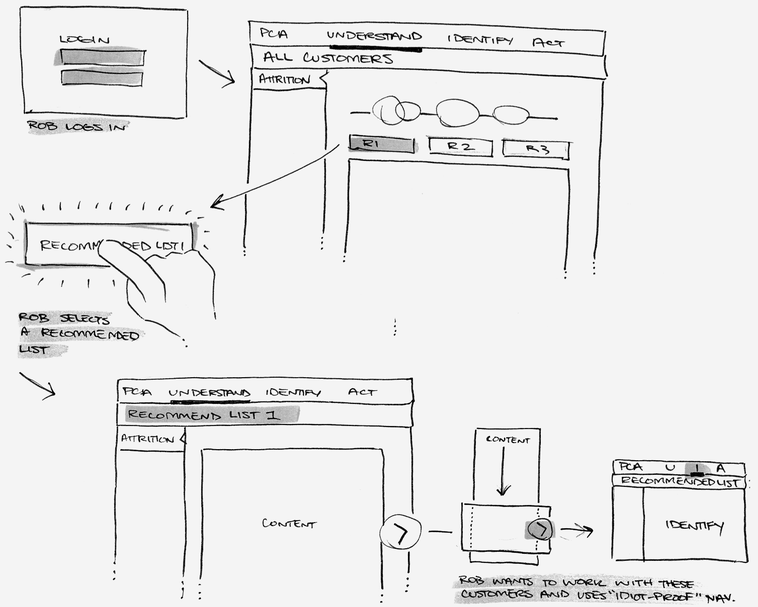
B

C



Mending ideas

A story-based exploration and a mid-fidelity wireframe show the workspace navigation taking shape. I phased out the remaining, clumsy global navigation (C) in final iterations.



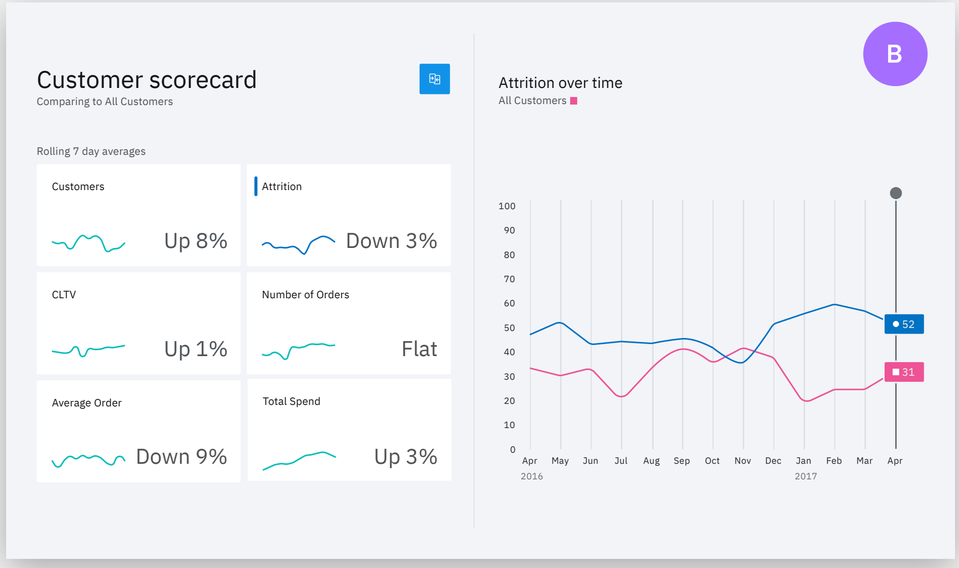
I explored a range of visualizations for the Customer Scorecard before settling

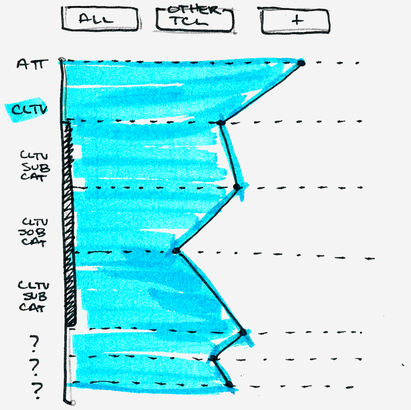
The Customer Scorecard enables marketing analysts to see a holistic picture of customers on a list, giving them useful context to ground their analysis in.

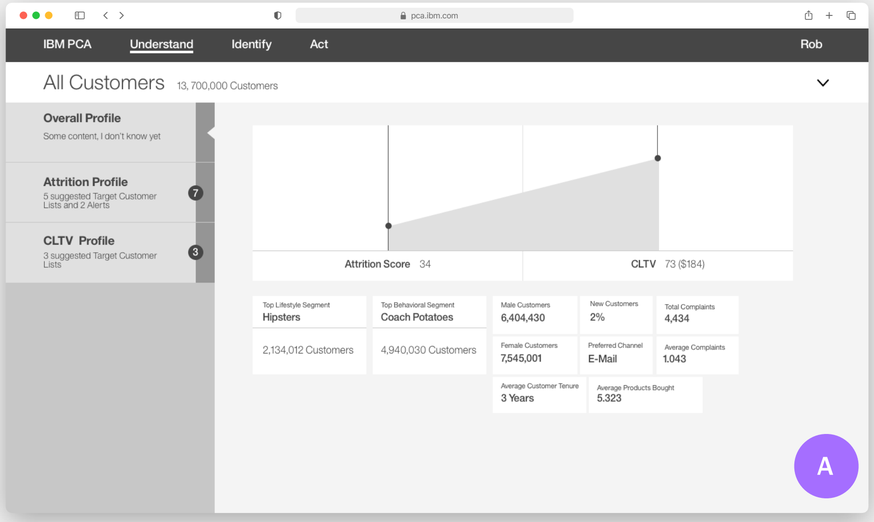
My role involved designing the layout and data visualizations of the scorecard. Initial wirefames explored a snapshot overview of customers with a multivariate visualization (A). But through evaluative research, the design evolved into a modular, time-based view that could be benchmarked with other customers (B).

Cards and sparklines

Customer attributes arranged as cards with sparklines employs the small multiples technique. A selected card unveils a line chart.

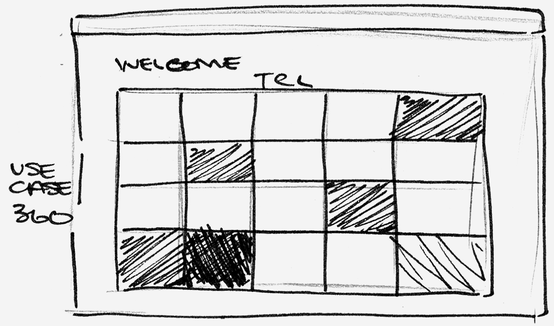






Parallel coordinates

Customer attributes scored and compared to one another create a multivariate index of a customer list.



Referencing heat maps

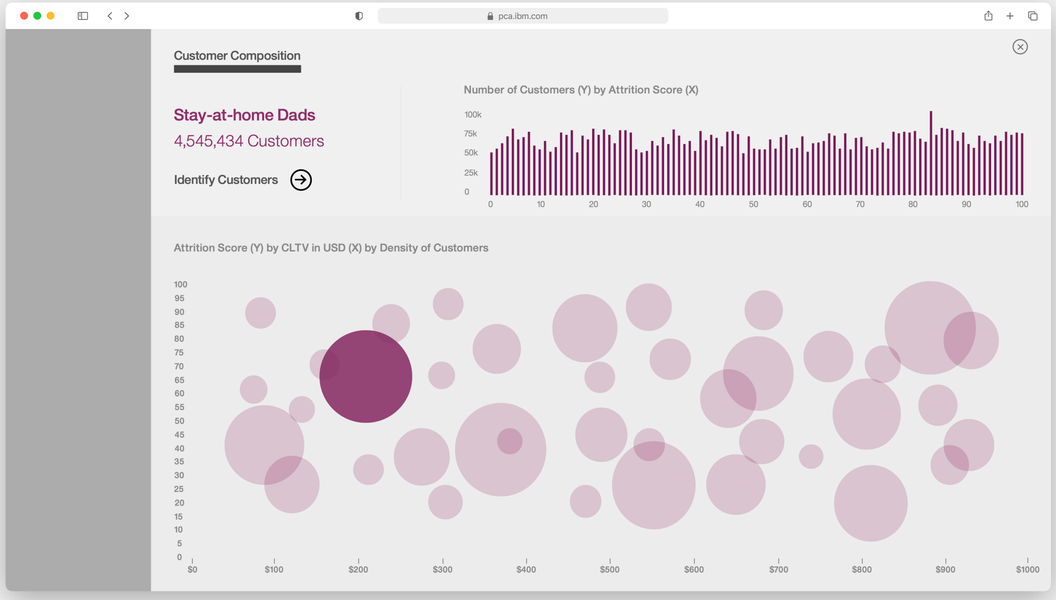
Customer attributes, such as CLTV, arranged on a grid with their statuses classified by color.

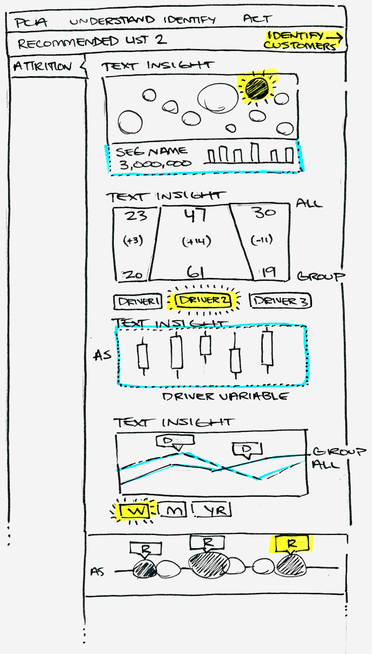
I negotiated the right data density for the Attrition Lens with offering management

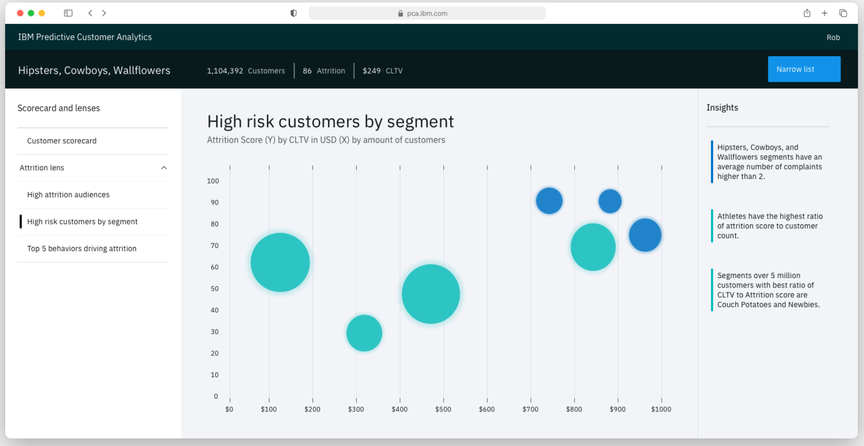
Lenses provide marketing analysts a view of customers through use cases such as retaining customers. Early designs were overloaded with data stemming from Offering Management's concern to provide insight (A).

I provided the voice to reduce the number of messages at a single time. As a result, charts are broken into individual views and insights are curtailed by prioritization.

A







Shifting elements

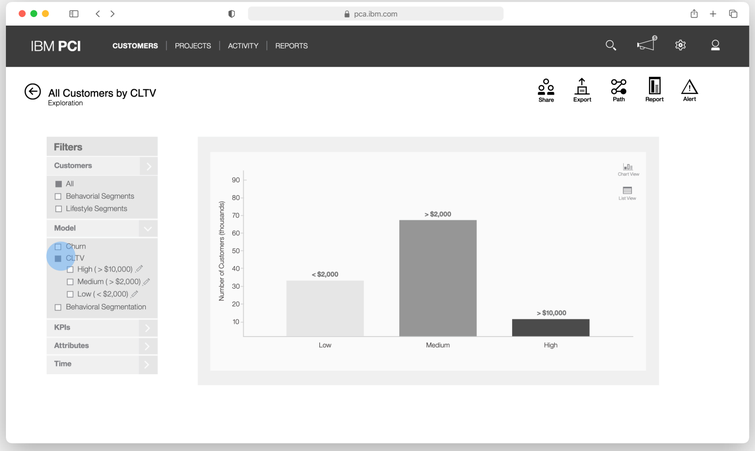
Lenses' messages moved from a long scrolling view to indexed in a menu. Insights moved from sitting on top of visualizations to positioned beside them. These decisions gave analysts more dexterity while reducing information overload.

By introducing narrative, I pushed past initial ideas about how to edit lists

Predictive analysis accelerated and enriched a marketing analyst's analysis, but it did not eliminate it. Therefore, a central challenge I had was helping analysts organize and control the decisions in their analysis.

In early designs, editing a list was filter-based (A). Problematically, this design overlooked preserving the narrative about an analysis. By later iterations I introduced a Rule Set component (B) that surfaces the path an analyst takes to get to a desired customer list.

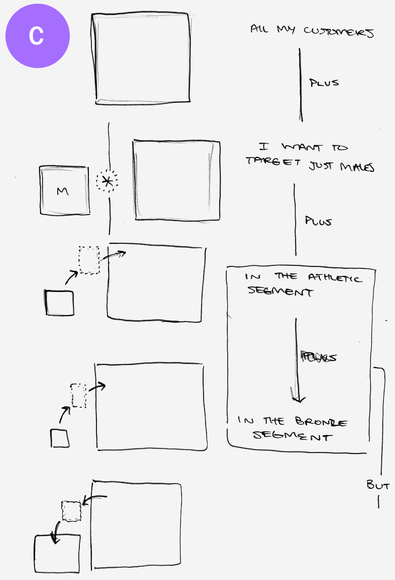
A



Filters with no narrative

This mid-fidelity prototype used in an early design review shows how marketing analysts narrowed a list using by toggling filters on and off.

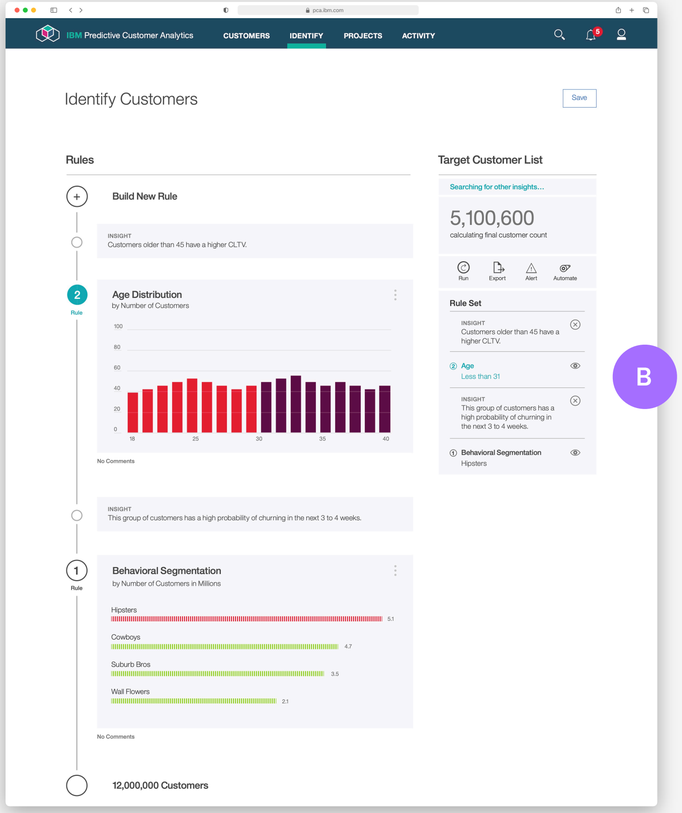
C



A visual rule builder emerges

In a major iteration, rules were arranged along a linear sequence to form a narrative about the list. The sketch shows the origin of the concept (C). Ultimately, the design's layout reached its limits with large amounts of rules and some rule making workflows. Many elements carried over into final iterations.

B

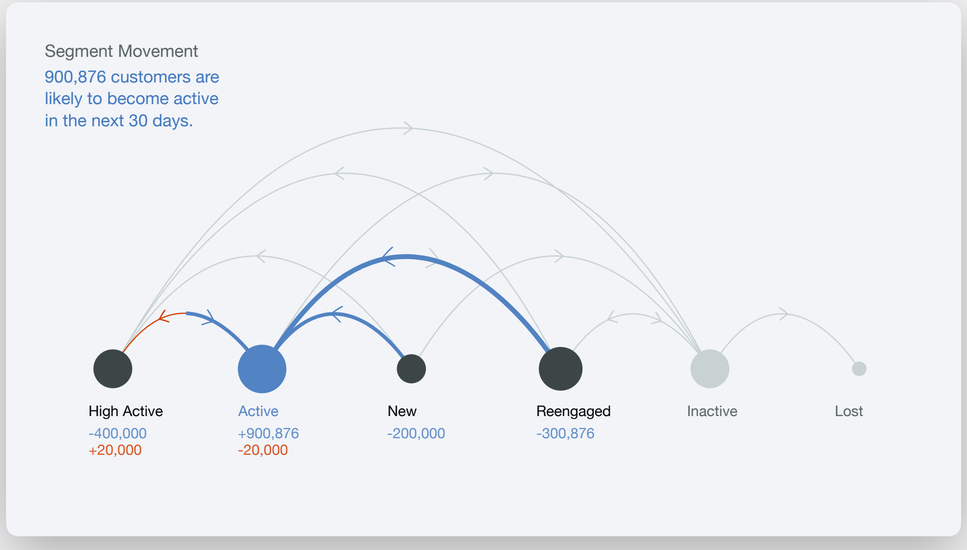


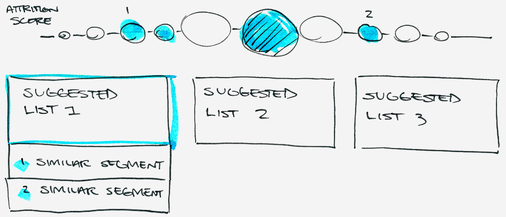
Bubbles represented customers and created a visual language for data

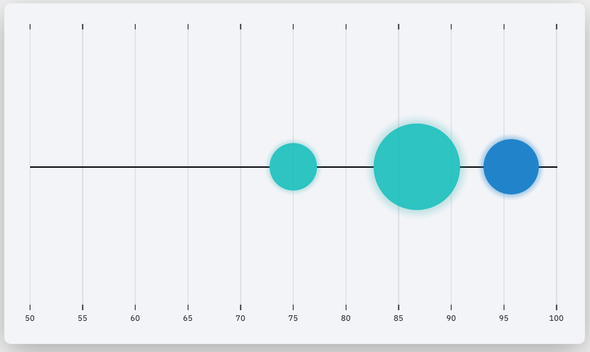
I selected data visualizations for the Attrition Lens and the for the future release of the Engagement Lens (A).

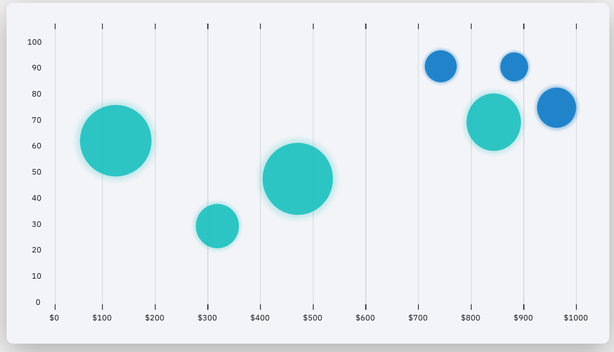
I selected visualizations that could represent customers as bubbles. Bubbles suggest people, engage the eye, and lend themselves to seen as interactive objects compared to other forms in data visualization. Used systemically, they create language distinct to PCA.

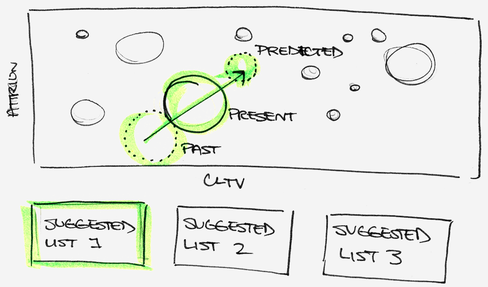
A

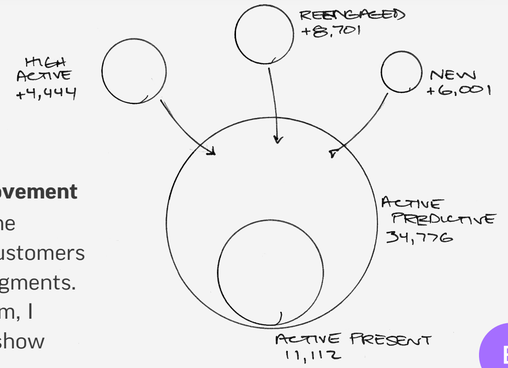












Visualizing customer movement

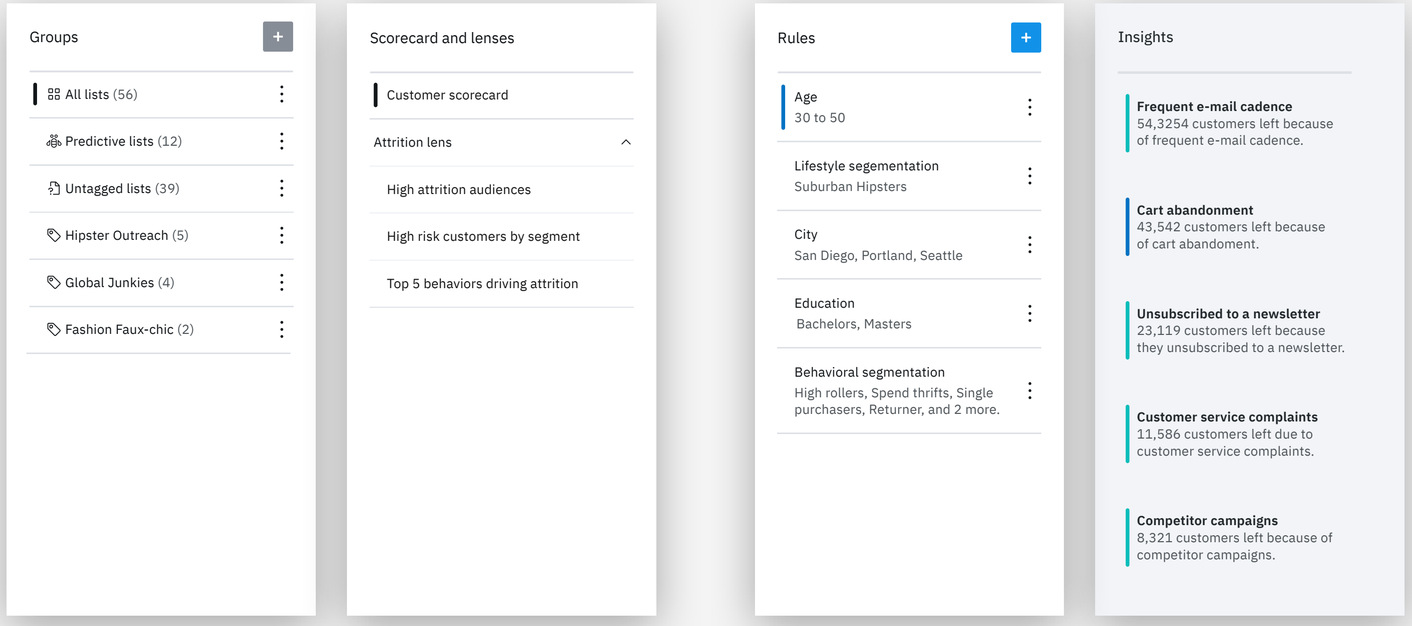
An arc diagram shows the complex movement of customers between engagement segments. Leading up to arc diagram, I explored other ways to show movement (B).

B

The Carbon Design System influenced how I designed the interface's components

Carbon, at the time, was a nascent design system with the goal of unifying the IBM product experience. During the final iteration I worked on PCA, the interface adopted elements of Carbon.

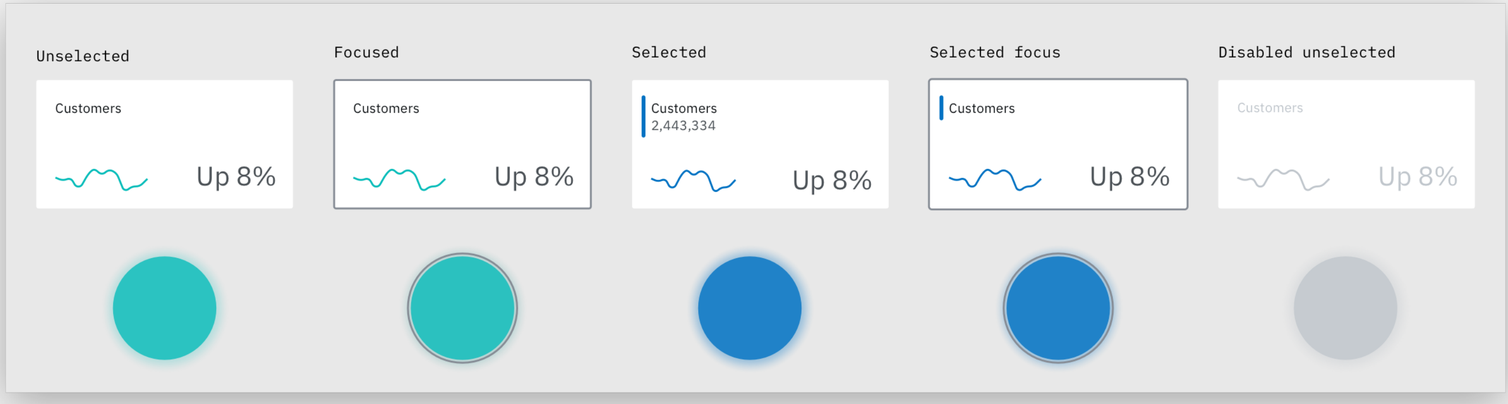
With Carbon as a foundation, I selected a color system and modified interface components to support the product's core workflows.



Navigation 1
One set of navigation colors represent movement through the application.

Navigation 2
A second set of navigation colors represent selecting and working with data.

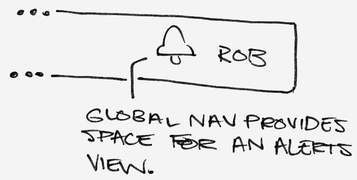
Consistency across elements
Teal functions as unselected data, cyan as selected data.

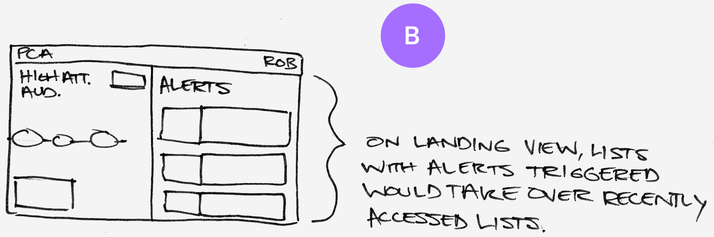


I designed the interface to consider how it could scale for future releases

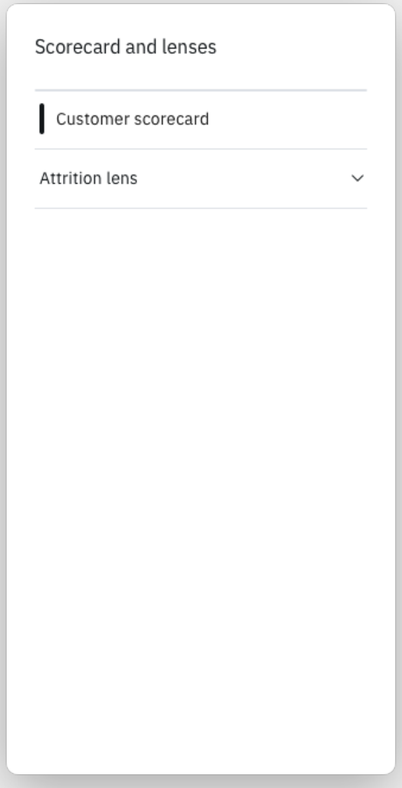
By the end of my time working on PCA, the product was released to a select set of clients. The product helped marketing analysts identify customers based on attrition, see a real-time understanding of customers, and allowed them to stream customer lists to marketing engagement systems.

Future experiences next in line for the product's roadmap were additional use case-driven lenses (A) and alerts (B). I considered these future experiences in the interface's initial release by how I arranged interface elements and in the application's map.

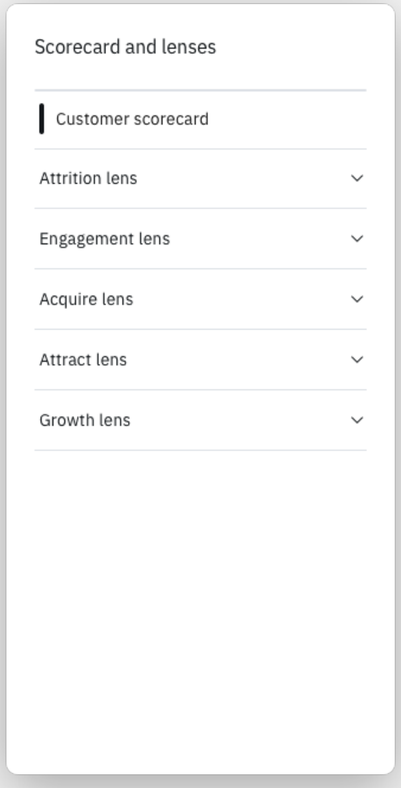




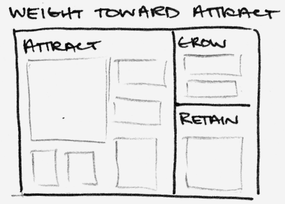
A

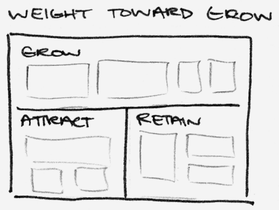


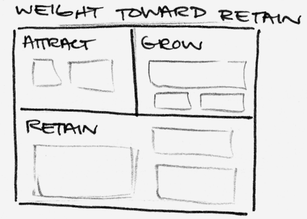
Current Understand navigation



Future Understand navigation







Impacts to landing view

Having multiple lenses would have implications for the types of predictive starting points PCA could offer marketing analysts. This sketch shows how user activity could prioritize what they see.

I brought PCA from its infancy to its limited release

While I had worked on a number of products before, PCA was the first product I made I major contributions towards. It is where I evolved from being a junior UX designer to a mid-level UX designer.

My role filled a void and provided form to the user experience. My contributions accelerated the project by driving alignment all the way from its initial market need framing to a product available to a limited set of clients.

PCA merges into a different product that IBM eventually sells

In early 2017, PCA integrated into an expanding and neighboring product called Watson Marketing Insights (WMI). In 2019, IBM sold WMI to Centerbridge Partners which formed a standalone marketing cloud company called Acoustic.

My message to my clients: you might be ***a*** user, but you are not ***the*** user.

Desiring to champion users, my career detoured toward research and facilitation

I can confidently look at anyone and tell them we could have made a more compelling product and arrived at important decisions faster if my leadership had prioritized the voices of users.

I felt so strongly about this that this project inspired me to take a detour in my design career. After PCA, I put visual design and user experience design on hold to round myself out with facilitation and design research skills.

PCA's conditions exposed my crude skills uncovering and articulating user needs

Working on this product was challenging. I was fresh in my design career, yet there was a vacuum of design leadership. In a year, I had 5 different design managers. The design team fluctuated in size and the most enduring teammate lasted only 2 quarters.

Through these circumstances, I emerged as the voice of design and gained respect among product management and development. But it also exposed shortcomings in my ability to validate design decisions and convince stakeholders to validate the assumptions we were making about users.

Most stakeholders agree users matter and also see themselves as the user

Focusing on users is worthwhile because it ensures technology works for people rather than people working for technology. This yields a pleasant experience and people expect that from software. Most stakeholders in technology and product management agree, but the complication is agreeing on how.

Some of PCA's stakeholders in technology and product management had marketing backgrounds and therefore identified as the user. And while it's true they could be *a* user and their perspective matters, they are not *the* user.

Seeing users as those with varying views and outside the team is advantageous

The user represents a group of people who are currently immersed in the work the product is for. It recognizes that not all users work or think the same and that past experience is distinct from current experience. By speaking with a group of as little as 5–7 users provides confidence that the product can handle a variety of contexts and perspectives users have.

It's also advantageous to select users outside the product team. Doing so insulates the team from the Curse of Knowledge. Outside users aren't familiar with the product and lack the mental models that form while working on the product that implicitly make it easier to understand.

Stakeholders can offload a designer's need for knowledge with users

Trusting design teams to access users takes effort to set up, but the effort pays a dividend. Designers often come onto projects knowing little about users and their problems. Kick off meetings just scratch the surface for the type of knowledge designers need to develop compelling solutions.

It's often unfeasible for stakeholders to spend with designers the time that they need to learn about the user's context and needs. However, stakeholders can distribute this load onto users. Speaking with 5 users for 2 one hour sessions over 2 weeks is much more manageable than spending that same time with a single stakeholder.